

Xianzheng Ma

Embodied Agents · World Action Models · 3D Understanding

DPhil Student, VGG & AVL, University of Oxford

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Research Interests

3D computer vision and robotics. Previously focused on leveraging 3D Large Language Models to reason about and understand the 3D world. More recently, shifted towards enabling robots to understand and interact with the physical world through video and 3D modalities, with a current emphasis on video-based world action models for cross-embodiment and sample-efficient robot learning.

Education

University of Oxford

DPhil, Visual Geometry Group (VGG) & Active Vision Lab (AVL)

Oct 2023 – Present

Oxford, UK

Supervised by [Prof. Victor Prisacariu](#) and [Prof. Iro Laina](#).

Working closely with [Prof. João F. Henriques](#) and [Prof. Ingmar Posner](#).

Wuhan University

Bachelor's and Master's Degrees

Sep 2014 – Jun 2021

Wuhan, China

Research Experience

Shanghai AI Laboratory

Full-time Researcher

2021 – 2023

Shanghai, China

Supervised by [Prof. Chao Dong](#).

Ongoing Projects

- **CompAct: compositional World and Video Action Models for robotics.** Building action-conditioned world models for robotic manipulation with large video diffusion backbones (Cosmos 2.5 and Wan 2.1), targeting compositional generalization and sample-efficient adaptation across task combinations, embodiments, and robot settings.
- **Language-guided self-evolving agents for physical video models.** Developing a Cosmos 3-based agentic post-training and evaluation loop that uses language critique, failure mining, and iterative data/model refinement to improve physical plausibility and temporal consistency.
- **RAG-based physical video modeling with Cosmos 3.** Building retrieval-conditioned generation methods that use relevant motion/video references to improve physical plausibility, long-horizon consistency, and robot-learning evaluation rollouts.

Selected Publications (* equal contribution)

1. Yongchang Zhang*, **Xianzheng Ma***, Tianyi Liu, Guangquan Zhou, Yang Chen. *See It, Say It, Sorted: An Iterative Training-Free Framework for Visually-Grounded Multimodal Reasoning in LVLMs*. **CVPR**, 2026. [\[Paper\]](#)
2. **Xianzheng Ma***, Tao Sun*, Shuai Chen, Yash Bhargat, Jindong Gu, Angel X Chang, Iro Armeni, Iro Laina, Songyou Peng, Victor Adrian Prisacariu. *Do 3D Large Language Models Really Understand 3D Spatial Relationships?*. **ICLR**, 2026; **CVPR 2026 3D-LLM/VLA Workshop Best Paper Runner-Up Award**. [\[Paper\]](#) [\[Project Page\]](#) [\[Code\]](#)
3. Guanqi Zhan*, **Xianzheng Ma***, Weidi Xie, Andrew Zisserman. *Inferring Dynamic Physical Properties from Video Foundation Models*. **CVPR 2026 VGBE Workshop Best Paper Runner-Up Award**. [\[Paper\]](#)

4. Kohsuke Ide, Ryosuke Yamada, Yue Qiu, **Xianzheng Ma**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Yutaka Satoh. *Beyond Single Object: Learning 3D Relations with Large Language Models*. **CVPR**, 2026 (Finding).
5. Yanbang Li, ZiYang Gong, Haoyang Li, Xiaoqi Huang, Haolan Kang, Guangpingbai, **Xianzheng Ma**. *Robotic Visual Instruction*. **CVPR**, 2025. [\[Paper\]](#) [\[Project Page\]](#)
6. **Xianzheng Ma***, Brandon Smart*, Yash Bhalgat*, et al.. *When LLMs Step into the 3D World: A Survey and Meta-Analysis of 3D Tasks via Multi-modal Large Language Models*. **IJCV**, under review. [\[Paper\]](#) [\[GitHub\]](#)
7. **Xianzheng Ma**, Zhixiang Wang, Yacheng Zhan, Yinqiang Zheng, Zheng Wang, Dengxin Dai, Chia-Wen Lin. *Both Style and Fog Matter: Cumulative Domain Adaptation for Semantic Foggy Scene Understanding*. **CVPR**, 2022 Oral. [\[Paper\]](#)
8. Min Shi, Zihao Huang, **Xianzheng Ma**, Xiaowei Hu, Zhiguo Cao. *Matching Is Not Enough: A Two-Stage Framework for Category-Agnostic Pose Estimation*. **CVPR**, 2023 Highlight. [\[Code\]](#)
9. Yulu Gan, **Xianzheng Ma**, Yan Bai, Yihang Lou, Renrui Zhang, Nian Shi, Lin Luo. *Decorate the Newcomers: Visual Domain Prompt for Continual Test Time Adaptation*. **AAAI**, 2023 **Outstanding Student Paper**. [\[Paper\]](#)
10. **Xianzheng Ma**, Hossein Rahmani, Zhipeng Fan, Bin Yang, Jun Chen, Jun Liu. *REMOTE: Reinforced Motion Transformation Network for Semi-supervised 2D Pose Estimation in Videos*. **AAAI**, 2022. [\[Paper\]](#)

Honors & Awards

- Best Paper Runner-Up Award, CVPR 2026 Workshop on Video Generative Models: Benchmarks and Evaluation (VGBE)
- Best Paper Runner-Up Award, CVPR 2026 3D-LLM/VLA Workshop
- Outstanding Reviewer (top 3%), ICCV 2025
- Outstanding Student Paper Award, AAAI 2023
- Oral Presentation, CVPR 2022
- Highlight Paper, CVPR 2023

Skills

Programming: Python, PyTorch, C/C++, CUDA, Linux, Git. Multi-node distributed training on GPU clusters (SLURM, PBS).

Robotics Simulation: MuJoCo, ManiSkill, Genesis.

Robotics & VLA: Pre-training and reproduction of VLA models (e.g., xVLA) with multi-node multi-GPU training, achieving 40× inference acceleration; reproduced and benchmarked on CALVIN, RoboTwin, SIMPLER, etc.

Learning from Human: Built a Real-to-Sim-to-Real pipeline that extracts manipulation skills from human demonstration videos and transfers them to robotic policies via simulation; teleoperation data collection on Franka Arm via Meta Quest and handheld controllers.

Academic Services

Conference Reviewer: CVPR (2022–2026), AAAI (2022–2026), ICLR (2023–2026), NeurIPS (2024–2025), ICCV (2025), ECCV (2024).

Journal Reviewer: IJCV, TPAMI, TMM.

Area Chair: IJCAI 2022 (China branch).